

Python: Secure Login System

Implement a Python program using closures to create a secure login system. The closure should:

- Store provided user credentials
- Allow access only when the correct username and password are entered
- Log each successful and failed login attempt

Function Description

Complete the function `login_system` in the editor with the following parameters:

- dict `login_attempts`: a dictionary with 'success' and 'failed' as keys and their counts as the values

Returns

- `function secure_login(string username, string password)`:
 - Manages the login process by checking the provided username and password against stored credentials.
 - Logs the attempt as either successful or failed.
 - Return Values:
 - If credentials are correct: "Welcome, {username}!"
 - If credentials are incorrect: "Failed login attempt for {username}"
- `function get_login_attempts()`:
 - This returns the `login_attempts` dictionary, which has keys 'success' and 'failed' to track the number of successful and failed login attempts.

Constraints

- The number of users is less than or equal to 30.

Input Format For Custom Testing

The integer in the first line denotes the number of users.

The following lines contain the {username}, {password} for each of the users.

The integer in the following line denotes the number of users trying to log in.

The following lines contain {username}, {password} for login attempts.

Sample Case 0

Sample Input For Custom Testing

```
4
admin,3rTb9ZaYqF6vP2dQ8wKmXsNcRJ5tL7hP2MqU
user1,1aXc8Yb0pR4eT9sD6uJnWqAcLJ3vK8eF1LpM
user2,5sFb4XnVqR2pT6dE9wKmSsNcZJ8tL5hP9MqU
user3,9aBc8Yd0pR4eT9sD6uJnWqAcLJ3vK8eF1LpM
2
admin,3rTb9ZaYqF6vP2dQ8wKmXsNcRJ5tL7hP2MqU
user1,1aXc8Yb0pR4eT9sD6uJnWqAcLJ3vK8eF1LpM
```

Sample Output

```
Welcome, admin!  
Welcome, user1!  
Login Attempts - Successful: 2 Failed: 0
```

Explanation

There are usernames and passwords for 4 users, and two login attempts. The credentials are correct for both users, so their welcome messages are displayed. Finally, the login attempt count is displayed.

Sample Case 1

Sample Input For Custom Testing

```
4  
admin,3rTb9ZaYqF6vP2dQ8wKmXsNcRJ5tL7hP2MqU  
user1,1aXc8Yb0pR4eT9sD6uJnWqAcLJ3vK8eF1LpM  
user2,5sFb4XnVqR2pT6dE9wKmSsNcZJ8tL5hP9MqU  
user3,9aBc8Yd0pR4eT9sD6uJnWqAcLJ3vK8eF1LpM  
1  
user,pass123
```

Sample Output

```
Failed login attempt for user  
Login Attempts - Successful: 0 Failed: 1
```

Explanation

There are credentials for 4 users and one failed login attempt.